[INCH-POUND]
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SUPERSEDING
MIL-G-80176B
15 November 1982

COMMERCIAL ITEM DESCRIPTION

GRINDING AND SANDING MACHINE, PLATEN, ABRASIVE BELT, WET AND DRY

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-G-80176B for all federal agencies.

1. SCOPE. This commercial item description (CID) covers wet and dry abrasive belt platen grinding and sanding machines and accessories intended for use in secondary finishing operations such as flat grinding, deburring, chamfering, rounding of edges, and freehand grinding.

2. CLASSIFICATION. The abrasive belt platen grinding and sanding machines shall be of the following types, classes, and sizes. The type, class, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Dry
Type II - Wet

Class A - Vertical platen
Class B - Horizontal platen
Class C - Combination vertical or horizontal platen
Class D - Combination adjustable vertical to horizontal platen

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3419

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TABLE I. Sizes of grinding and sanding machines.

Size	Belt width, inches
1	1 (type I only)
2	2
3	3
4	4
6	6
8	8
10	10
12	12
14	14

3. SALIENT CHARACTERISTICS

3.1 General requirements. The grinding or sanding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The abrasive belt grinding or sanding machine shall include, as a minimum, the following components.

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine.

3.2.2 Frame. The frame shall have provisions for mounting the hood and side guards, platen, abrasive belt drive system, abrasive belt, and workpiece support.

3.2.3 Hood and side guards. Unless otherwise specified (see 7.2(c)), the machine shall have a hood and side guards to minimize the discharge of abrasive dust, abraded metal, and water spray toward the operator.

3.2.4 Platen. Unless otherwise specified (see 7.2(d)), the platen shall provide a work area length that will be a minimum of one and one-half times the width of the belt. Unless otherwise specified (see 7.2(e)), the platen shall be constructed of the material considered by the manufacturer as standard.

3.2.5 Workpiece support. Unless otherwise specified (see 7.2(f)), the grinding machine shall have a worktable or other workpiece support constructed of the material considered by the manufacturer as standard.

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3.2.6 Abrasive belt system. The abrasive belt system shall consist of abrasive belt contact wheels and an electric motor with either a direct drive coupling or a belt drive system. The abrasive belt system shall have means for adjusting the tension and tracking alignment of the abrasive belt. When a motor belt drive system is utilized, a means for adjusting the tension of the drive belts shall be provided.

3.2.7 Abrasive belt. If required, the grinding machine shall be furnished with abrasive belts of the quantity, grit type, and grit size as specified (see 7.2(g)). Belt sizes shall be in accordance with International Organization for Standardization (ISO) 1929, "Abrasive belts – Dimensions, Tolerances and Designation".

3.2.8 Manual controls. Unless otherwise specified (see 7.2(h)), the machine shall be provided with the manufacture's standard machine controls.

3.2.9 Coolant system. Type II grinding or sanding machines shall have a coolant system. Unless otherwise specified (see 7.2(i)), the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping.

3.2.10 Electrical system. Unless otherwise specified (see 7.2(j)), the electrical system shall conform to the American National Standards Institute/National Fire Prevention Association (ANSI/NFPA) 79, "Electrical Standard for Industrial Machinery". Motors less than one horsepower shall have 115-volt, 1-phase, 60-hertz circuit capability. Motors one horsepower or greater shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(k)).

3.2.11 Electric motors. The belt grinding machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.2(l)), the motors shall be totally enclosed and fan cooled (TEFC).

3.2.12 Optional accessories. Optional accessories shall be provided as specified (see 7.2(m)).

3.3 Performance. Unless otherwise specified (see 7.2(n)), the belt grinding or sanding machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, and adjusting mechanisms shall be verified during the trial period.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(o)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910,"Occupational Safety and Health Standards for General Industry".

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3.6 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(p)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.7 Lubrication plate or chart. When specified (see 7.2(q)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with Society of Automotive Engineers (SAE) number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Warranty. Unless otherwise specified (see 7.2(r)), the manufacturer's standard commercial warranty, terms, and conditions shall apply.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the machine shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(s)).

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7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ISO standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Type, class, and size of grinding and sanding machine required (see 2)
c. Hood and side guards, if different (see 3.2.3)
d. Platen size, if different (see 3.2.4)
e. Platen material, if different (see 3.2.4)
f. Workpiece support, if different (see 3.2.5)
g. Abrasive belts, as specified (see 3.2.7)
h. Manual controls, if different (see 3.2.8)
i. Coolant system, if different (see 3.2.9)
j. Electrical system industrial standard, if different (see 3.2.10)
k. Electrical system voltage, as specified (see 3.2.10)
l. Electric motors, if different (see 3.2.11)
m. Optional accessories, as specified (see 3.2.12)
n. Performance, if different (see 3.3)
o. Machine dimensions and weight, as specified (see 3.4)
p. Nameplate, if different (see 3.7)
q. Lubrication plate, if required (see 3.8)
r. Warranty, if different (see 5.2)
s. Packaging requirements, if different (see 6.1)

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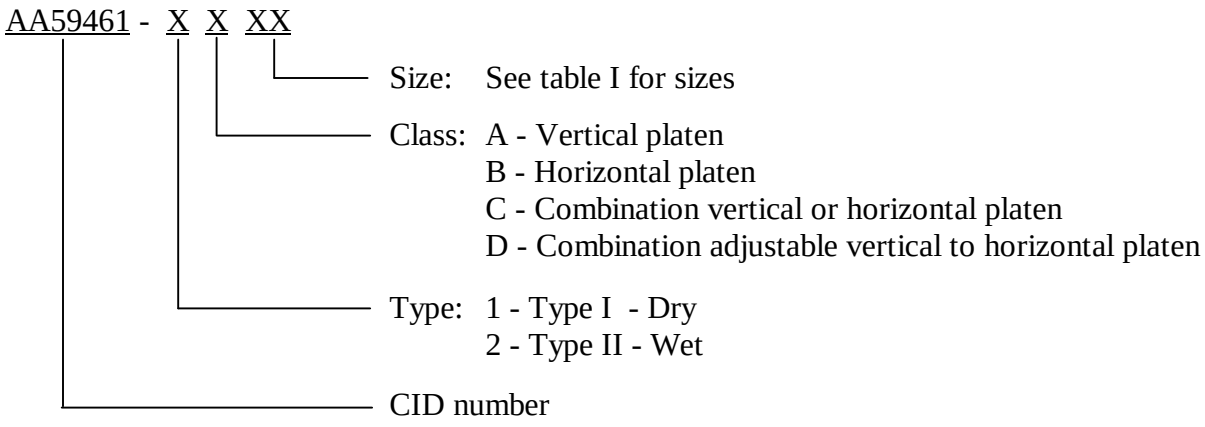
7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians:
Army - AL
Navy - SH
Air Force - 99

Reviewers:
Navy - AS, MC, OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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